INCH-POUND
MIL-PRF-49450/15(CR)
10 March 2000

PERFORMANCE SPECIFICATION SHEET

BATTERY, RECHARGEABLE, NICKEL-CADMIUM, VENTED, AIRCRAFT, BB-664/A

This specification is approved for use by the USACECOM, Department of Army, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-49450 (CR)

Performance.

Voltage. The voltage of the battery shall be 24 Vdc (nominal).

Capacity. The battery capacity shall be 13 ampere hours.

Weight. The weight of the battery shall be 27 pounds (Max).

Connector(J1). The main power connector shall be a quick disconnect type that meets the dimensional and interface requirements of MS3509.

Connector(J2). The connector for the electrical harness shall be a 10 pin, connector that meets the interface requirements of MS27468T11B35P.

Cells. The battery shall consist of 19 BB-475/A cells in series.

AMSC N/A: FSC 6140

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

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DRAWING NOTES:

All dimensions are in inches, unless specified otherwise.

Main power receptacle (J1). The battery shall utilize a two-pin receptacle conforming to the interface requirements of MS3509. The J1 receptacle shall be centered as shown in figure 1.

Sensor receptacle. (J2). The battery shall use a hermetic, jam nut receptacle, which mates with the connector MS27468T11B35P. The J2 receptacle shall be centered as shown in figure 2.

Bottom flange mounting. The battery shall have bottom flange on each side of the battery as shown in figure 1 and figure 2.

Battery container. The body and cover shall be made of the same alkali-resistant material. The dimensions of the container and lid shall conform to figure 1, figure 2 and figure 3.

Container venting. The battery shall provide a 0.06" gap between the battery case and cover for ventilation.

Harness assembly. The harness assembly shall conform to the applicable specification sheet.

Cell voltage tap. The battery shall provide a connection to the positive terminal of the tenth cell referenced to the battery negative. The voltage measured from this tap to the battery negative terminal shall be 5.26 $\pm$ 1.5 % of the overall battery voltage during open circuit charged conditions.

Color. The color of the container and cover except latches, hold-down hooks, other external hardware, pads, and identification markings and identification shall conform to FED-STD-595 within the range of blue color chips 15090 to 15193.

REQUIREMENTS:

Nomenclature:	BB-664/A Battery, Rechargeable
Rated capacity:	13 ampere-hours
Voltage (Nominal):	24 Volts
Weight (Maximum):	27 pounds
Connector:	one 2 pin connector that meets the dimensional and interface requirements of MS3509(J1)
Connector:	one 10 pin connector that meets the dimensional and interface requirements of MS27468T11B35P(J2)

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Test	Discharge Rate(Amps)	Cut Off Voltage(V)	Capacity(AH) Minimum
Capacity	8.7	18	13
Cycle Life			
Normal Rate	13	18	11.7
High Rate	130	14.4	8.5
Vibration	13	18	13
Temperature Shock	13	18	13
Altitude	13	18	11
Humidity and Charge Retention	13	18	9.75
High Rate			
Discharge and Low Temperature	130	13	6.5
High Rate			
Discharge and Operating Position	130	14.4	8.5
Shock	13	18	13

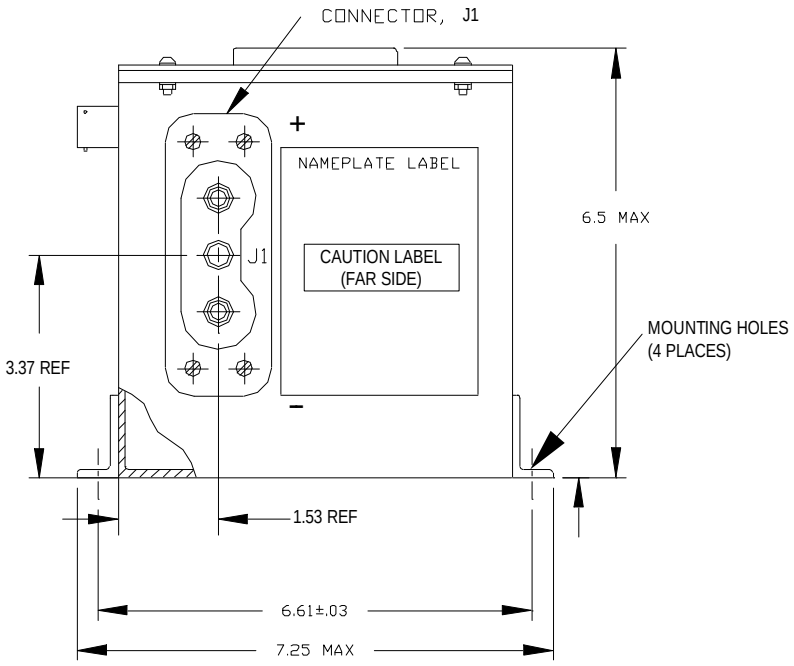

FIGURE 1: BATTERY RECHARGEABLE BB-664/A DIMENSIONS AND FRONT VIEW

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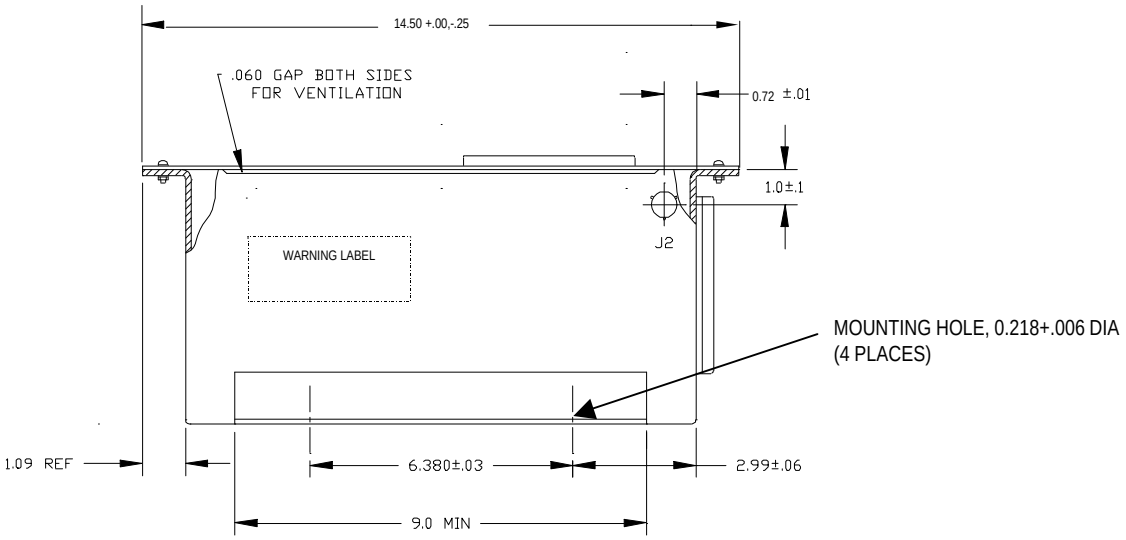

FIGURE 2: BATTERY RECHARGEABLE BB-664/A DIMENSIONS AND SIDE VIEW

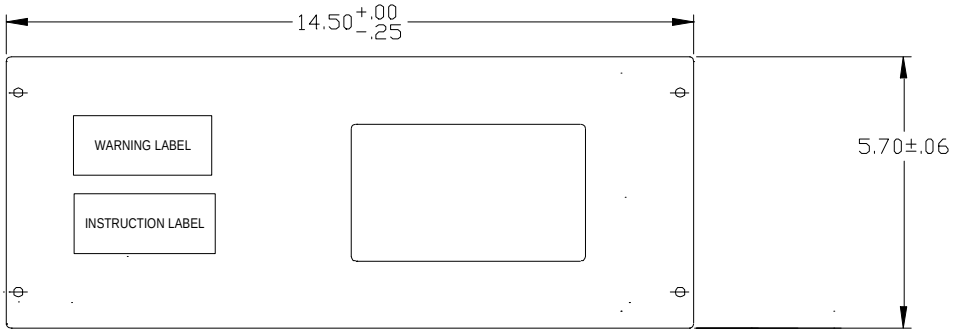

FIGURE 3: BATTERY RECHARGEABLE BB-664/A COVER TOP VIEW

Custodian:
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